# quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

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## Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain analytically siloed because each was processed with a different software configuration. We present `quantmsdiann`, an open-source Nextflow/nf-core workflow that parallelises `DIA-NN` across cloud and HPC infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for any `DIA-NN` version through Docker and Singularity, requiring no manual installation or dependency resolution, and natively handles all vendor formats, converting to

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mzML when needed (e.g. SCIEX). It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a `pmultiqc` quality-control report. The parallelisation design allows a 2,300-run single-cell dataset to be reanalysed in 2.4 hours on an HPC cluster of 300 nodes.

We benchmark `quantmsdiann` on the full ProteoBench DIA panel (five `DIA-NN` versions × four instrument modalities), reanalyse three independent bulk cell-line cohorts (PXD003539, PXD004701, PXD030304), and assemble a pan-cohort of five public DIA cell-line reanalyses under a single SDRF-driven invocation; SDRF-driven reprocessing with current `DIA-NN` versions matches or exceeds the originally deposited results. Single-cell and spatial proteomics demonstrations are part of ongoing work. `quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann`.

*Keywords:* DIA-NN, single-cell proteomics, spatial proteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultiqc

## 1. Introduction

Data-independent acquisition (DIA) is now the dominant mass-spectrometry strategy for proteomics at scale, and `DIA-NN` [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm `DIA-NN`'s competitiveness for single-cell DIA workflows [4], and versions 1.9.x and 2.x have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. Despite this analytical maturity, most `DIA-NN` users run the tool as a desktop application, which limits its scalability for large datasets. Moreover, deployments on HPC and cloud infrastructure remain artisanal, relying on shell scripts, ad-hoc directory conventions, and hand-curated sample sheets, so the same raw data analysed by two laboratories, or with two `DIA-NN` releases, can yield non-overlapping quantification tables that are difficult to reconcile.

Beyond per-study analysis, the value of public proteomics archives such as PRIDE Archive [5] lies as much in systematic reanalysis as in original deposition. In 2024 we published bigbio/quantms [6], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [7] input on the nf-core framework [8]. quantms was, however, built around DIA-NN 1.8.1 with a fixed parameter set, and its DDA/DIA breadth made extending DIA-specific functionality difficult: new DIA-NN parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run DIA-NN invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of nf-core, the sensitivity of current DIA-NN releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present quantmsdiann, an independent SDRF-driven Nextflow pipeline dedicated to DIA-NN that emerged from the need to extend DIA-specific features beyond what the original bigbio/quantms pipeline could accommodate. While quantmsdiann shares its SDRF-first design philosophy and parts of the configuration-generation tooling with quantms, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every DIA-NN version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard DIA-NN reports and MSstats input, bundling the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival. We demonstrate the pipeline along three axes: ProteoBench [9] benchmarks across four DIA modalities and five DIA-NN releases; three independent bulk-DIA cell-line reanalyses (PXD003539, PXD004701, PXD030304); and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation.

## 2. Experimental Procedures

### *2.1. Pipeline implementation*

quantmsdiann is implemented in Nextflow DSL2 ($\geq$25.10.4) and follows nf-core community standards [8] for pipeline structure, testing, documentation, and continuous integration. Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer); upstream community tools (DIA-NN 1.8.1, ThermoRawFileParser [10], pridepy [11], qpx) are pulled from BioContainers, while pipeline-specific images (newer DIA-NN releases and wiffconverter) are hosted on ghcr.io/bigbio. The repository at https://github.com/bigbio/quantmsdiann is organised as a standard nf-core pipeline with main workflow (main.nf), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.1.0 (release codename “Sao Pablo”, 2026-05-08) was used for the analyses reported in this paper.

### *2.2. SDRF parsing and configuration generation*

`quantmsdiann` accepts SDRF [7] as primary input. SDRF parsing is performed using the sdrf-pipelines Python package developed within the quantms project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. `DIA-NN` configuration generation extracts enzyme, fixed and variable modification, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings from the SDRF and writes a `DIA-NN` command-line configuration. Where SDRF columns are absent, `DIA-NN` defaults appropriate for the inferred instrument are applied.

### *2.3. Raw file handling and `DIA-NN` execution*

Raw inputs are dispatched to format-specific handlers in the file-preparation subworkflow, each conversion or decompression running as an independent, parallelised Nextflow process. Thermo `.raw` files are read natively by `DIA-NN` $\geq$ 2.1.0 (which added Linux native-reader support) or converted to mzML with ThermoRawFileParser [10] otherwise (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (`https://github.com/bigbio/wiffconverter`), removing the historical dependency on a proprietary ProteoWizard installation. Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`). `DIA-NN` [1] is then invoked at the version chosen via `-diann_version`: the ProteoBench analyses use `DIA-NN` 1.8.1, `DIA-NN` 2.1.0, `DIA-NN` 2.2.0, `DIA-NN` 2.3.2, and `DIA-NN` 2.5.0 from the same SDRF, and the cell-line reanalyses use `DIA-NN` 2.5.0 unless otherwise stated.

### *2.4. Spectral-library generation*

`quantmsdiann` runs `DIA-NN` library-free, generating the spectral library in silico from the input FASTA rather than supplying an external experimental library: `DIA-NN`’s deep-learning predictor builds a predicted library from the protein sequences, each run is searched against it, a cohort-wide empirical library is assembled from the confident first-pass identifications, and a final pass re-searches and quantifies against that refined library. The source FASTA was the UniProt human reference proteome [12] (UP000005640, *[release]*), with match-between-runs enabled at the cohort level. In ProteoBench's submission taxonomy this is the predicted-library strategy (`predictors_library` = DI-ANN); the ProteoBench comparison (Sec. 3.3) is therefore restricted to community submissions that used the same predicted-from-FASTA strategy.

### *2.5. Quality control and QPX export*

Per-run and per-cohort quality control reports are generated by `pmultiqc` [13], a proteomics extension of MultiQC [14] that summarises `DIA-NN`-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (`https://github.com/bigbio/qpx`), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [15] downstream analysis.

### *2.6. Datasets benchmarked*

The benchmark and reanalysis datasets used in this study (ProteoBench DIA modules 5/7/9/10, the three per-cohort cell-line reanalyses PXD003539, PXD004701, PXD030304, the two additional pan-cohort datasets PXD041421 and PXD017199, and the PXD071075 scalability sweep) are summarised together with their instrument, acquisition mode, and deposition year in Additional file 1: Table S1. For each dataset, an SDRF was prepared from the original sample metadata and is provided as part of the Additional file 1 deposition. For each cell-line reanalysis cohort, the original published quantification was downloaded from the PRIDE deposition and used as the comparator, thresholded at 1% precursor-level FDR for both `quantmsdiann` and the original analyses. The conservative contaminant/decoy filter and the like-for-like-threshold rule applied to the original-paper headline counts are described in Additional file 1.

### *2.7. Compute environment and benchmarking*

Runtime benchmarks were performed on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory allocations dispatched by Nextflow. Memory profiling was performed using the Nextflow trace report (`-with-trace`).

## 3. Results

### *3.1. End-to-end throughput on archive-scale DIA cohorts*

The headline operational result of `quantmsdiann` is that a single SDRF-driven invocation processes archive-scale DIA cohorts within hours of wall-clock time on commodity HPC, with wall-clock time decreasing monotonically with cluster width over the tested range up to a practical knee near 200 nodes. On the PXD071075 single-cell DIA benchmark (~2,300 raw files, Wang *et al.* [4]), increasing the Nextflow `executor.queueSize` from 10 to 300 cluster nodes reduced end-to-end wall-clock time from 37.7 h to 2.4 h (Fig. 1b). Scaling was monotonic across the sweep, with a practical knee near 200 nodes (2.6 h) beyond which the cohort-level consolidation step dominates and additional parallel workers yield diminishing returns. Beyond the 200-node knee, gains in wall-clock time come at decreasing CPU-hour efficiency: scaling from 10 to 300 nodes reduces wall-clock by 15.5× for a 30× increase in cluster width, so users should choose cluster width based on the desired turn-around time rather than the maximum-throughput operating point. Across ProteoBench benchmark datasets and our public cell-line cohorts, final-run wall-clock time tracked cohort size and instrument family without per-dataset workflow changes (Fig. 1c): the largest cohort in the panel, PXD030304 (5,798 TripleTOF 6600 runs), completed in 4.8 h on the same configuration that finishes the 6-run ProteoBench cohorts in under one hour. Peak memory usage during the consolidation step remained below *[GB]* GB for cohorts up to *[Nmax raw files]* raw files, allowing `quantmsdiann` to be deployed on commodity 64–128 GB HPC nodes rather than requiring high-memory specialist hardware. Per-step wall-clock and resource envelopes are reported in Additional file 1: Figs. S1 and S2.

### *3.2.* `quantmsdiann` *pipeline architecture*

`quantmsdiann` is implemented in Nextflow DSL2 [8] and follows nf-core community standards, distributing each computational step as a versioned Docker or Singularity/Apptainer container. The pipeline accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. From this input, `quantmsdiann` derives the analysis parameters required by `DIA-NN` automatically, eliminating the most common source of analysis-script divergence between research groups. The workflow comprises four mandatory and three optional modules organised around the

DIA-NN analysis core (Fig. 1a), alternating between per-file parallel stages (red; in-silico library generation when requested, preliminary calibration, individual search) and collective stages (green; empirical library assembly, final quantification, MSstats conversion, pmultiqc [13] quality-control reporting) operating on the full cohort.

#### *3.2.1. Parallelisation strategy*

Raw-file conversion and per-run quantification are dispatched as independent Nextflow tasks, enabling the throughput scaling reported in Section 3.1. Dataset-wide DIA-NN steps that require the full cohort (match-between-runs, normalisation, protein inference) run in a single consolidation task with tuned memory ceilings, avoiding out-of-memory failures common to single-machine DIA-NN runs on 1,000+ file inputs. Outputs include the standard DIA-NN report tables (`report.tsv`, `report.pr_matrix.tsv`, `report.pg_matrix.tsv`) alongside MSstats-formatted input and a pmultiqc [13] quality-control report. A harmonised QPX archive bundles the SDRF, identifications (PSM, peptide, protein-group, feature), MS metadata, ontology terms, and pipeline provenance into a Parquet-based artefact, which can be queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [15] downstream analysis.

#### *3.2.2. DIA-NN version flexibility*

Container-pinned profiles for DIA-NN 1.8.1, 2.1.0, 2.2.0, 2.3.2 and 2.5.0 are shipped with the pipeline, with the DIA-NN binary selected at runtime through the `-diann_version` parameter; users can add further versions by supplying a custom container. Identical SDRF inputs can therefore be processed against multiple DIA-NN versions in one invocation, with separate output directories per version. This explicit version handling enables cross-version reproducibility audits without re-engineering the workflow and supports use cases such as reanalysis of historical datasets with the DIA-NN version originally used by the depositing laboratory, alongside re-quantification with the current release.

#### *3.2.3. Library handling*

For single-cell and spatial DIA experiments, matched empirical spectral libraries from bulk samples are often unavailable. quantmsdiann supports two complementary library modes: (i) DIA-NN library-free analysis from a user-supplied FASTA proteome (default), in which DIA-NN generates a predicted in-silico library and a cohort-wide empirical library is then assembled across all runs in the SDRF; and (ii) user-supplied spectral library passed via `-speclib`, for users wishing to reuse a previously generated empirical or hybrid library. In-silico library generation, per-run calibration, and empirical-library assembly are each parallelised across the relevant runs.

### *3.3. ProteoBench benchmark across `DIA-NN` versions and instruments*

To demonstrate that `quantmsdiann` preserves `DIA-NN`'s analytical performance under SDRF-driven orchestration, we processed every public ProteoBench DIA module through the workflow and compared the resulting identification counts against the ProteoBench public-submission snapshot (29 May 2026) [9]. `quantmsdiann` runs `DIA-NN` library-free: it predicts the spectral library in-silico from the FASTA and refines it from the acquired data, with no externally supplied experimental library. For a like-for-like comparison we therefore restricted the community set to ProteoBench `DIA-NN` submissions that used the same predicted-from-FASTA library (`predictors_library` = DIANN); submissions that loaded a pre-existing experimental or user-curated spectral library search a different candidate space and are excluded from the headline comparison. The benchmark panel covers four instrument families and acquisition modes: Orbitrap Astral 2 Th DIA (*Module 7*), single-cell DIA on Orbitrap (*Module 9*, PXD049412), timsTOF diaPASEF (*Module 5*, PXD062685), and SCIEX ZenoTOF 7600 (*Module 10*, PXD070049). For each module we ran the workflow against five `DIA-NN` releases (1.8.1, 2.1.0, 2.2.0, 2.3.2, and 2.5.0) in a single invocation, with all five versions parametrised from the same SDRF (Fig. 2).

Identification counts grew monotonically with `DIA-NN` version on every module: precursors rose by 14% on Module 7 (105,793 $\rightarrow$ 120,973 from 1.8.1 to 2.5.0), 14% on Module 9 (38,218 $\rightarrow$ 43,761), 27% on Module 5 (91,075 $\rightarrow$ 115,735), and 17% on Module 10 (85,847 $\rightarrow$ 100,524); protein-group counts at 1% FDR rose in lockstep (7% / 8% / 14% / 7% respectively). This monotonic version-progress trend is consistent with what the broader ProteoBench community reports across `DIA-NN` releases [9], confirming that the SDRF wrapper introduces no version-coupling artefacts. On the two modules for which the snapshot contained predicted-library `DIA-NN` submissions—Orbitrap Astral (Module 7, $n = 15$) and timsTOF diaPASEF (Module 5, $n = 8$)—`quantmsdiann` sits within the predicted-library community distribution in the identification-versus-accuracy plane (Fig. 2b), at precursor counts and median quantification error $|\epsilon|$ comparable to the leading submissions; parameter-matched per-submission comparisons are provided in Additional file 1: Fig. S3. The remaining two modules (single-cell Module 9 and ZenoTOF Module 10) had no predicted-library `DIA-NN` submission in the snapshot, so their panels show the `quantmsdiann` version trajectory alone. Although identification counts rise monotonically with `DIA-NN` version (Fig. 2a), median quantification error $|\epsilon|$ does not always improve with the newest release: on a per-module basis the best identification–accuracy trade-off may sit at an intermediate `DIA-NN` version rather than the most recent one (Fig. 2b).

### *3.4. Independent cell-line reanalyses recover comparable or deeper coverage than the original quantifications*

To evaluate `quantmsdiann` on production-scale public deposits, we reanalysed three independent bulk-DIA cell-line cohorts that had been processed and published with different DIA engines (Fig. 3). For each cohort the same raw files were retrieved from PRIDE Archive [5], parsed through an SDRF generated from the original sample annotations, and run through `quantmsdiann` with current `DIA-NN` releases.

On the Guo *et al.*, 2019 NCI-60 panel (PXD003539; originally analysed with OpenSWATH using the Guo 2019 assay library), `quantmsdiann` recovered 95,633 peptides at 1% FDR against 40,592 in the original submission (+135%), with comparable protein-group counts (6,747 versus 6,556; Fig. 3a). Because the original Guo 2019 analysis used OpenSWATH with a fixed assay library while `quantmsdiann` runs `DIA-NN` in library-free mode, this gain reflects both the SDRF-driven orchestration and the change of analysis engine; a like-for-like `DIA-NN`-on-original-library control is part of ongoing work. On the Sun *et al.*, 2023 breast-cancer panel of 76 cell lines (PXD004701; originally analysed with PCT-SWATH), `quantmsdiann` recovered 6,146 protein groups under the original consistency filter (proteotypic peptides, $\leq$90% missing) against 6,091 in the published analysis (+0.9%), with raw-matrix counts within 10% on the unfiltered metrics (Fig. 3b). On the ProCan-DepMapSanger 949-cell-line pan-cancer panel (PXD030304; originally analysed with `DIA-NN` 1.7.x), `quantmsdiann` recovered 8,921 protein groups at 1% FDR against 8,498 originally (+5%) and 7,990 protein groups requiring $\geq 2$ unique peptides versus 6,692 (+19%; Fig. 3c). Both the original 8,498 protein-group headline (Global.Q.Value $\leq 0.01$, no per-protein peptide threshold) and the stringent 6,692 count (same Global.Q.Value filter plus a $\geq 2$ supporting-peptide requirement) are reported in the ProCan-DepMapSanger paper itself; the `quantmsdiann` numbers (8,921 and 7,990) are computed under the matching filters and the same conservative contaminant/decoy filter, so each side of the comparison is like-for-like. Across the three cohorts, the gain was largest on the deposits that had been processed with older or non-`DIA-NN` engines (Guo 2019 OpenSWATH, ProCan 1.7.x); the cohort that had already been analysed with a current DIA engine (Sun 2023) yielded near-parity, as expected. Per-condition completeness, peptide-per-protein distributions, gene-versus-accession overlap and per-cell-line missingness for each cohort are provided as Additional file 1: Figs. S5–S10.

### *3.5. Pan-cohort of five public DIA cell-line reanalyses*

To illustrate the value of running the same SDRF-driven workflow across many independent deposits, we combined the three reanalysed cohorts above with two additional cell-line DIA panels (PXD041421, Mehnert 2023; PXD017199, Tognetti 2021, 67 breast cancer cell lines) into a single `quantmsdiann` invocation and assembled a pan-cohort view of cell-line and protein-accession overlap (Fig. 4). The combined panel comprises 12,229 unique UniProt accessions across the five cohorts, of which 5,144 (42.1%) are detected in all five, defining a pan-cohort core proteome (Fig. 4a, upper UpSet plot); a further 1,334 (10.9%) are shared by exactly four of the five cohorts. Per-cohort headline counts (lower-left bars in Fig. 4a) confirm that `quantmsdiann` matches or exceeds the original published count on each cohort.

Tissue-level coverage across the combined panel spans 27 Cellosaurus-derived tissue categories (plus the catch-all "Other tissue" and "Healthy (Non-cancer)" bins; Fig. 4b): lung, breast, haematopoietic-and-lymphoid, skin, and central nervous system together account for 59% of all cell-line entries in the pan-cohort. Because every count comes from running a single SDRF-parameterised `quantmsdiann` invocation rather than reconciling per-study quantifications, the resulting protein matrix is directly queryable from the published QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation. The consolidated SDRF and per-cohort `quantmsdiann` configurations underpinning this pan-cohort run will be deposited on Zenodo before publication, so that the full input–configuration–output triple can be re-executed by third parties.

## 4. Discussion

`quantmsdiann` lowers the engineering barrier between a deposited DIA dataset and a reproducible quantification table, and it does so at a scale that

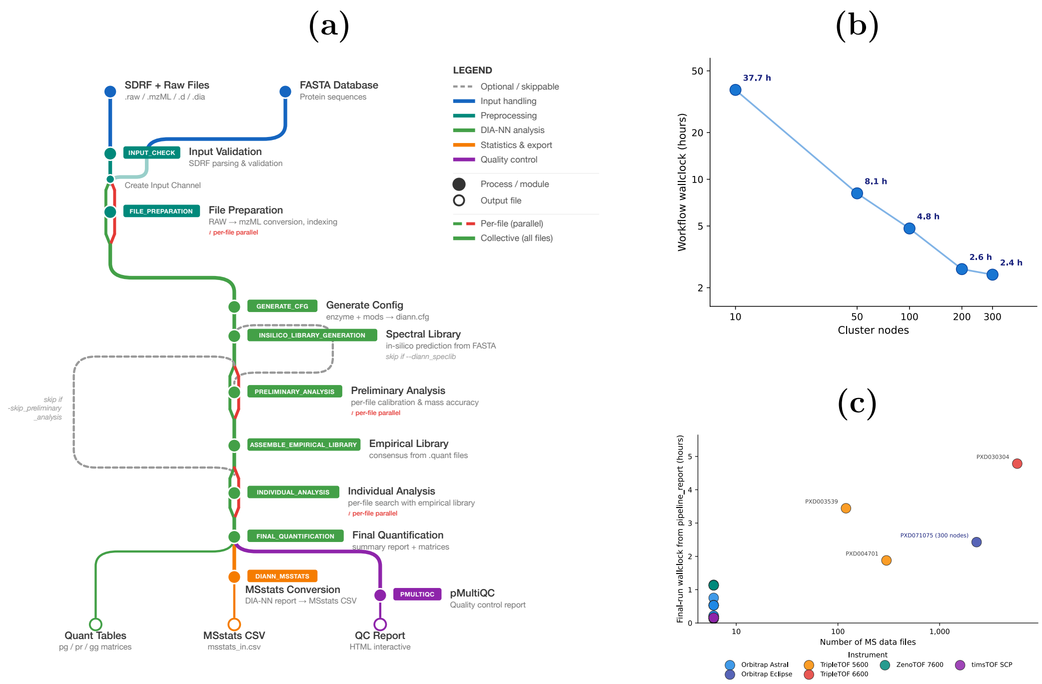


Figure 1: `quantmsdiann` **pipeline architecture and runtime scaling.** (**a**) Subway diagram of the `quantmsdiann` workflow. Mandatory modules are shown as filled circles, optional/skippable modules as open circles; per-file parallel stages are marked in red, collective cohort stages in green. The pipeline consumes an SDRF and a FASTA reference, optionally converts vendor raw files to mzML, generates an in-silico or empirical spectral library, runs per-file `DIA-NN` analyses in parallel, consolidates results in a single cohort-level final quantification, and emits standard `DIA-NN` tables, an MSstats input, a QPX Parquet archive, and a `pmultiqc` quality-control report. (**b**) Workflow wall-clock time versus Nextflow `executor.queueSize` (log-scaled cluster nodes; ticks at 10, 50, 100, 200, 300) on the PXD071075 single-cell cohort (Wang *et al.*, 2025); wall-clock drops from 37.7 h at 10 nodes to 2.4 h at 300 nodes, with the practical knee near 200 nodes. (**c**) Final-run wall-clock versus number of MS data files across ProteoBench benchmark and cell-line cohorts, coloured by instrument family; the PXD071075 production run (300 nodes) is annotated.

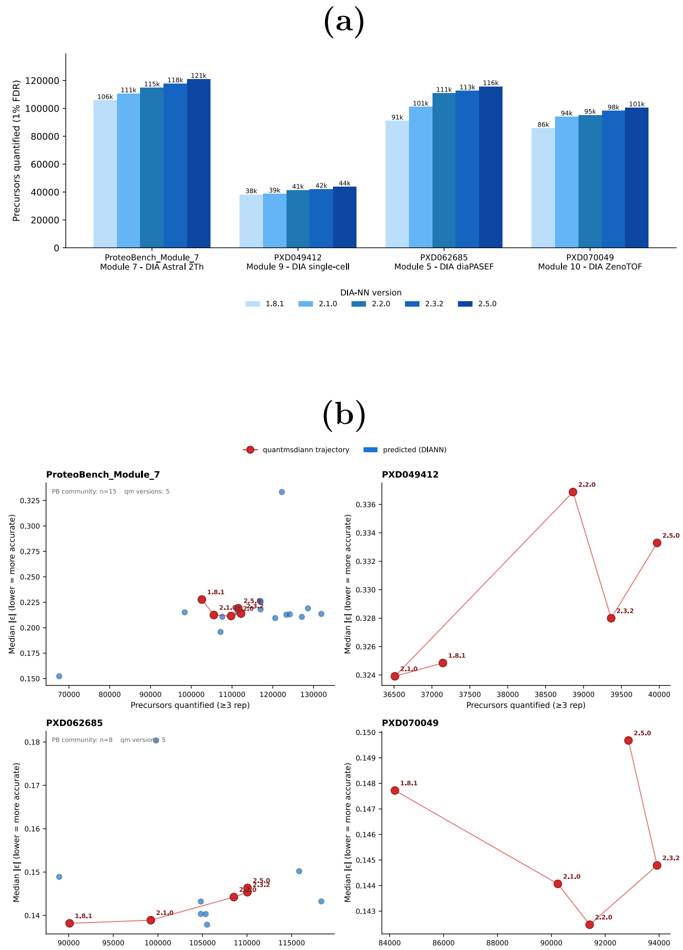



Figure 2: **quantmsdiann on ProteoBench across five DIA-NN releases and four DIA modalities.** Four ProteoBench DIA modules cover Orbitrap Astral (Module 7), single-cell DIA (Module 9, PXD049412), timsTOF diaPASEF (Module 5, PXD062685) and SCIEX ZenoTOF (Module 10, PXD070049). (**a**) Precursors at 1% FDR by DIA-NN version (1.8.1–2.5.0; light to dark blue) from one quantmsdiann invocation; identification counts grow monotonically with version (e.g. Module 5 91,075 → 115,735). (**b**) Precursors quantified ($\geq$3 replicates) versus median $|\epsilon|$ for the ProteoBench DIA-NN submissions that used a predicted-from-FASTA library (predictors_library = DIANN; snapshot 29 May 2026) — the apples-to-apples set matching quantmsdiann's library-free strategy; the quantmsdiann trajectory across the five DIA-NN versions is overlaid in red. The single-cell (Module 9) and ZenoTOF (Module 10) panels carried no predicted-library DIA-NN submission in the snapshot and so show the quantmsdiann trajectory alone. Unlike the identification-count trend in panel (a), median $|\epsilon|$ does not always decrease with the newest DIA-NN version. See Additional file 1: Fig. S3 for parameter-matched per-submission comparisons.

**(a) PXD003539 — Guo 2019, NCI-60** **(b) PXD004701 — Sun 2023, breast cancer** **(c) PXD030304 — ProCan, 949 lines**

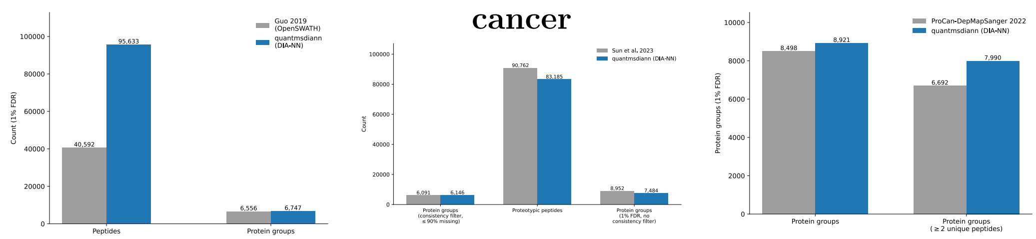
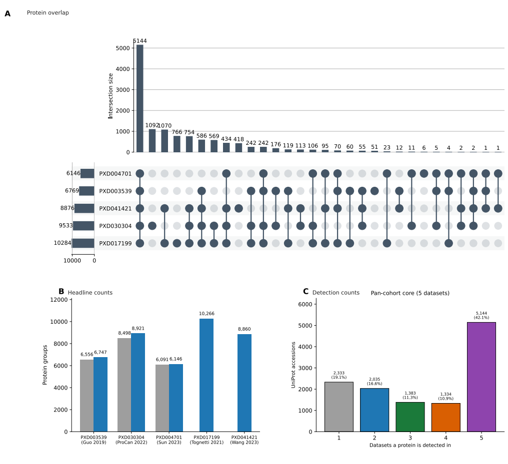



Figure 3: **Independent reanalysis of three public cell-line DIA datasets recovers as many or more identifications than the originally deposited quantifications.** Each panel compares the originally published peptide and/or protein-group counts (grey) against the `quantmsdiann` reanalysis at 1% FDR (blue) using the same raw data. (**a**) Guo *et al.*, 2019 NCI-60 OpenSWATH analysis (PXD003539, original engine = OpenSWATH with the Guo 2019 assay library): `quantmsdiann` recovers 95,633 peptides versus 40,592 originally (+135%) and a comparable 6,747 protein groups versus 6,556. Because `quantmsdiann` runs `DIA-NN` in library-free mode while the original ran OpenSWATH against a fixed assay library, this gain reflects both SDRF-driven orchestration and the change of analysis engine; a like-for-like `DIA-NN`-on-original-library control is part of ongoing work. (**b**) Sun *et al.*, 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 cell lines): under the original consistency filter (–90% missing, proteotypic) `quantmsdiann` recovers 6,146 versus 6,091 protein groups (+0.9%); the raw matrix counts are within 10% on both proteotypic-peptide and protein-group metrics. (**c**) ProCan-DepMapSanger 2022 pan-cancer cell-line panel (PXD030304, 949 lines): `quantmsdiann` recovers 8,921 protein groups at 1% FDR versus 8,498 originally (+5%), and 7,990 versus 6,692 after requiring $\geq 2$ unique peptides per protein (+19%). Both the original 8,498 headline (Global.Q.Value $\leq 0.01$) and the stringent 6,692 count (Global.Q.Value $\leq 0.01$ plus $\geq 2$ supporting peptides) are reported in the ProCan-DepMapSanger paper itself; the `quantmsdiann` counts are computed under the matching filters, so each side of the comparison is like-for-like. Per-condition completeness, peptide-per-protein distributions and gene- versus accession-level overlap are shown in Additional file 1: Figs. S5–S10.

**(a) Protein-accession overlap across cohorts**



**(b) Per-tissue coverage**

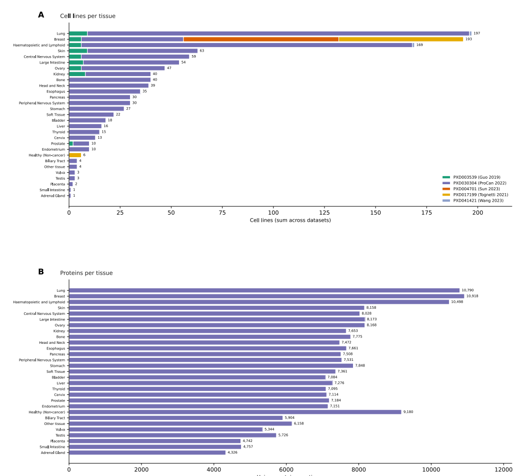


Figure 4: **Pan-cohort of five public DIA cell-line reanalyses by `quantmsdiann`.** Combines PXD003539 (NCI-60, Guo 2019), PXD030304 (ProCan-DepMapSanger, 949 lines), PXD004701 (Sun 2023, 76 breast cancer lines), PXD041421 (Mehnert 2023), and PXD017199 (Tognetti 2021, 67 breast cancer lines), reanalysed under a single `quantmsdiann` invocation. (**a**) UpSet plot of pan-cohort UniProt-accession overlap with per-cohort headline counts: a 5,144-protein pan-cohort core is detected across all five cohorts. (**b**) Per-tissue distribution of cell lines (top) and unique proteins (bottom) across the combined panel.

makes systematic reanalysis of public DIA archives tractable for the first time. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is interoperable with the wider proteomics ecosystem [7]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container so that reanalysed results can be archived (e.g. on Zenodo) and re-queried alongside the original raw data. By focusing exclusively on `DIA-NN`, rather than the multi-engine scope of the related bigbio/quantms pipeline [6], `quantmsdiann` achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file `DIA-NN` invocation rather than across-engine comparison.

The demonstrations we present stress the workflow along three orthogonal axes. The ProteoBench benchmark (Fig. 2) tests analytical fidelity across four instrument modalities and five `DIA-NN` releases; the three independent cell-line reanalyses (Fig. 3) test recovery against published per-study quantifications, with results ranging from near-parity (Sun 2023, current DIA engine) to substantial recovery gains (Guo 2019 OpenSWATH; ProCan `DIA-NN` 1.7.x); and the pan-cohort of five public DIA cell-line reanalyses (Fig. 4) tests archive-scale throughput on five concurrently reanalysed deposits. Single-cell and spatial proteomics biology demonstrations are part of ongoing work and are not part of this submission's biology figures; the single-cell PXD071075 cohort used as the throughput-scaling benchmark in Fig. 1 is, however, real production data. This pattern argues that the value of DIA reanalysis is real but time-limited—largest for deposits originally analysed with older or non-`DIA-NN` engines and shrinking toward parity for recently processed cohorts—exactly the regime an SDRF-driven workflow is well-positioned to address as new `DIA-NN` versions ship. Demonstrations on flagship single-cell DIA and laser-microdissected spatial cohorts are part of the development roadmap; the workflow already supports these acquisition modes (Fig. 1a, library-handling branch) and SDRFs for the Brunner [2], Derks [16], Leduc [17], Wang [4], Mund [18] and Makhmut [19] datasets are in preparation.

The cross-`DIA-NN`-version reproducibility view enabled by `quantmsdiann` is, to our knowledge, the first pipeline-level treatment of this question for `DIA-NN`. We expect this to become increasingly important as `DIA-NN` evolves: a deposited single-cell DIA dataset analysed with `DIA-NN` 1.8 today and reanalysed with `DIA-NN` 2.x in two years should not produce different biological conclusions for the same data and parameters, but in current practice such re-analyses are rarely performed because the bespoke per-study scripts make them prohibitively expensive. `quantmsdiann`'s container-pinned, version-parametrised execution makes this kind of audit a single-command operation.

Several limitations are worth noting. First, `quantmsdiann` does not currently include batch correction, cell-type assignment, or differential-abundance modules; these are best handled in downstream R or Python packages such as scp [20], which can read the QPX or MSstats outputs directly. Second, library generation for plexDIA channel deconvolution remains internal to `DIA-NN`, and `quantmsdiann` inherits any limitations of that step. Third, the QPX format will continue to evolve as community feedback accumulates; backward compatibility is maintained through schema versioning. Fourth, we focused on bulk DIA cell-line reanalyses; plexDIA, TMT-DIA, and single-cell DIA demonstrations are part of the development roadmap but are not evaluated here.

Finally, we note that `quantmsdiann` is complementary to, rather than competing with, existing single-cell DIA workflows. The benchmarking work of Wang *et al.* [4] compared multiple search engines and rescoring strategies; `quantmsdiann` adopts the broadly competitive `DIA-NN` engine and focuses on the orthogonal problem of orchestrating that engine reproducibly at scale. Similarly, earlier nf-core DIA efforts such as DIAproteomics [21] and, more recently, the MHCquant2 pipeline [22] address analogous orchestration problems for DIA and immunopeptidomics. Together, these contributions extend the nf-core proteomics ecosystem to cover the major MS-based proteomics modalities behind a common SDRF-driven interface.

## Declarations

*Ethics approval and consent to participate*

Not applicable. All datasets reanalysed in this study were obtained from public proteomics repositories with ethics approval and informed consent from the original depositing studies.

*Consent for publication*

Not applicable.

*Data availability*

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Mehnert 2023). Scalability experiment: PRIDE PXD071075 (Wang 2025). ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Additional file 1: Table S1. Per-process runtime and resource envelopes (Figs. S1, S2) and detailed ProteoBench supplementary panels (Figs. S3, S4) are provided as Additional file 1.

*Code availability*

`quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann` under the MIT license; documentation is hosted at `https://quantmsdiann.quantms.org/`. The pipeline is implemented in Nextflow DSL2 ($\geq$25.10.4) and packages each step as a versioned Docker, Singularity, or Apptainer container image. Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at `https://github.com/bigbio/quantmsdiann-paper` (this repository).

*Competing interests*

*[VD is the original author of DIA-NN. Other authors declare no competing interests.]*

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*Authors' contributions*

*[QXY, YS and CD developed the pipeline. AL and HW implemented the SDRF parsing and QPX export. JN and FC contributed the spatial-proteomics datasets and analysis. VD provided DIA-NN expertise and guidance on cross-version reproducibility. MB and YPR conceived and supervised the project. QXY and YPR wrote the manuscript with input from all authors. All authors read and approved the final manuscript.]*

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